

UNF

Sales Order

Fiction Company
202 N. Main
Manhattan, KS 66502

CustomerNumber:	<i>1001</i>	Sales Order Number:	<i>405</i>
Customer Name:	<i>ABC Company</i>	Sales Order Date:	<i>2/1/2000</i>
Customer Address:	<i>100 Points</i>	Clerk Number:	<i>210</i>
	<i>Manhattan, KS 66502</i>	Clerk Name:	<i>Martin Lawrence</i>

Item Ordered	Description	Quantity	Unit Price	Total
800	widgit small	40	60.00	2,400.00
801	tingimajigger	20	20.00	400.00
805	thingibob	10	100.00	1,000.00
Order Total				3,800.00

First normal form

CusNum	CusName	CusAdd	SalesOrNum	SalesOrDate	ClerkNum	ClerkName	ItemNum	ItemName	Quantity	UnitPrice	Total	OrderTotal
1001	ABC Comp	100 Points Manhattan, KS 66502	405	2/1/2000	210	Martin Lawrence	800	Widget small	40	60	2400	3800
1001	ABC Comp	100 Points Manhattan, KS 66502	405	2/1/2000	210	Martin Lawrence	801	tingimajigger	20	20	400	3800
1001	ABC Comp	100 Points Manhattan, KS 66502	405	2/1/2000	210	Martin Lawrence	805	thingibob	10	100	1000	3800

Identify PK: CusNum, SalesOrNum, ItemNum, ClerkNum

Second Normal Form

Customer

CusNum	CusName	CusAdd
1001	ABC Comp	100 Points Manhattan, KS 66502

Item

ItemNum	ItemName	UnitPrice
800	Widget small	60
801	tingimajigger	20
805	thingibob	100

Clerk

ClerkNum	ClerkName
210	Martin Lawrence

SalesOrder

SalesOrNum	CusNum	ClerkNum	SalesOrDate	OrderTotal
405	1001	210	2/1/2000	3800

SalesOrderDetails

SalesOrNum	ItemNum	Quantity	Total
405	800	40	2400
405	801	20	400
405	805	10	1000

Third Normal Form

Identify transitive dependency, is there any? No, therefore the tables are in 3NF

*Some examples may not have transitive dependency, same goes to full dependency, some examples may not have full dependency.

Third Normal Form

Customer

CusNum	CusName	CusAdd
1001	ABC Comp	100 Points Manhattan, KS 66502

Item

ItemNum	ItemName	UnitPrice
800	Widget small	60
801	tingimajigger	20
805	thingibob	100

Clerk

ClerkNum	ClerkName
210	Martin Lawrence

SalesOrder

SalesOrNum	CusNum	ClerkNum	SalesOrDate	OrderTotal
405	1001	210	2/1/2000	3800

SalesOrderDetails

SalesOrNum	ItemNum	Quantity	Total
405	800	40	2400
405	801	20	400
405	805	10	1000

Normalization process ends, now can implement those tables in the DBMS.

Further changes can be made (optional) such as introducing FK / PK like below

Tables Implemented

Customer

CusNum	CusName	CusAdd
1001	ABC Comp	100 Points Manhattan, KS 66502

Item

ItemNum	ItemName	UnitPrice
800	Widget small	60
801	tingimajigger	20
805	thingibob	100

Clerk

ClerkNum	ClerkName
210	Martin Lawrence

SalesOrder

SalesOrNum	CusNum (FK)	ClerkNum (FK)	SalesOrDate	OrderTotal
405	1001	210	2/1/2000	3800

SalesOrderDetails

ID	SalesOrNum (FK)	ItemNum (FK)	Quantity	Total
001	405	800	40	2400
002	405	801	20	400
003	405	805	10	1000